

# LAU WEI XIANG

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## EDUCATION

### Singapore Management University

Bachelor of Science (Information System)

Aug 2024 - Apr 2028

- GPA: 3.99/4.0 (Summa Cum Laude, Highest Distinction)
- SMU Merit Scholarship, SMU SCIS Dean's List AY 2024/2025

## EXPERIENCE

### PCA Company Services

Automation Engineer Intern

Dec 2025 – Jan 2026

- Architected a full-stack booking system using FastAPI and React.ts, reducing processing time by 40% through the implementation of RESTful API endpoints.
- Engineered an end-to-end content automation pipeline leveraging Python, Google Apps Script and Gemini to WordPress, cutting manual article generation and publication time by 80% for 5 weekly articles.
- Reduced manual reporting effort by 80% by developing a Python-based desktop application that automated the generation of payroll reports and analytics dashboards from Excel data.
- Deployed and managed containerized applications with Docker on Google Cloud Run.

### Singapore Management University

Teaching Assistant

Jun 2025 – Dec 2025

- Reduced manual administrative overhead by 75% by developing an automated class participation tracking script using Python.
- Guided 40+ students in deploying and managing virtual servers on AWS EC2, enhancing practical understanding of cloud infrastructure and system configuration.

## PROJECTS

### Bidly

Project Lead

Jan 2026 – Present

- Architected a auction platform using microservices pattern with FastAPI, enabling independent scaling.
- Implemented asynchronous task processing using RabbitMQ, decoupling microservices and improving system responsiveness during peak auction traffic.
- Developed a WebSocket service to push live auction updates, reducing frontend polling and improving real-time bid visibility.
- Deployed microservices using Docker, orchestrated with Kubernetes achieving a scalable, self-healing, and load-balanced containerized system across nodes for high availability.
- Integrated Stripe and Twilio to handle payments and notification for secure transactions and timely alerts.

### SMUxChange

Project Lead

Aug 2025 – Dec 2025

- Led the development of a web application for SMU students to discover and plan exchange programs, reducing planning time by 50%.
- Built a responsive, modern frontend using React.js, Tailwind CSS, Bootstrap, and Shadcn UI, ensuring seamless cross-device usability, visual consistency and intuitive user experience.
- Directed and implemented backend development with Node.js integrating multiple API endpoints and managing Firebase for database operations and user authentication.

## CO-CURRICULAR ACTIVITIES

### Coders Assembly

Mentor

Aug 2025 – Dec 2025

- Mentored 7 freshman with no prior programming experience in Python fundamentals, introducing core concepts and best practices, with 100% of mentees achieving A- and above

## TECHNICAL SKILLS

- Programming Languages:** Python, Java, JavaScript, PHP, HTML, CSS, SQL
- Frameworks:** FastAPI, Flask, React.js, Vue.js, Node.js, Tailwind CSS, Bootstrap, Pandas
- Tools:** Docker, Kubernetes, RabbitMQ, Git, WebSocket
- Cloud:** AWS, Azure, Firebase, Vercel, Google Cloud